

VMware Virtual Lab Environment Setup

Project summary

Built a reusable virtualized Windows environment to support Active Directory, identity administration, PowerShell, and cybersecurity labs. This foundational infrastructure was used for subsequent domain and IAM exercises.

Lab environment

Component	Configuration
Host OS	Windows 11
Virtualization	VMware Workstation Player 17
Host RAM	16 GB (current; upgraded during later lab work)
Virtual machines	Windows Server 2022; Windows 11 client
Virtual network	NAT / VMnet8; 192.168.116.0/24
Later domain controller	JMC-DC01 (configured in subsequent AD lab)

Objectives

- Install and configure a desktop virtualization platform.
- Create Windows Server and Windows client virtual machines.
- Configure NAT networking for communication between lab systems.
- Troubleshoot network configuration and prepare for domain deployment.

Implementation overview

- Set up VMware Player on the Windows 11 host.
- Prepared Windows Server 2022 and Windows 11 guest environments.
- Used VMware NAT (VMnet8) networking and worked with virtual adapter settings.
- Investigated addressing, DHCP and connectivity issues while preparing the environment.
- Reused the environment for later Active Directory and IAM projects.

Validation and results

The environment supported later Windows Server domain configuration and Active Directory exercises. This report summarizes the setup; original VM configuration and connectivity screenshots have not been included or independently revalidated here.

Skills demonstrated

Virtualization | Windows Server | Windows 11 | NAT networking | Infrastructure troubleshooting | Lab documentation

Portfolio evidence to add

- VMware library showing both virtual machines (without private host details).
- VMnet8/NAT configuration or guest IP configuration.
- A successful VM connectivity check, if available.

Note: Private lab network details are shown for technical context. No passwords, secrets, or access tokens should be published.